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Amritapuri Campus

Course: 22ENV231 – Environmental Management

MOSAIC-Air

A Wind-Aware Spatiotemporal Deep Learning Framework for Urban Air Pollution Forecasting and Emerging Hotspot Detection

Project Team

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Abstract

Air pollution poses a persistent environmental and public-health challenge, particularly in rapidly urbanising regions where pollutant concentrations vary with meteorological conditions, emission patterns, and local geography. Although recent advances in deep learning have improved air-quality forecasting, several challenges remain, including sparse monitoring networks, dynamic spatial dependencies, unreliable detection of high-pollution episodes, and limited integration of predictive models into accessible environmental decision-support systems. This project proposes MOSAIC-Air, a wind-aware spatiotemporal intelligence framework for short-term air pollution forecasting and emerging hotspot detection. The proposed Wind-Aware Graph–Temporal Fusion Network (WAGTF-Net) integrates graph attention for inter-station relationships, temporal convolution and attention for time-dependent pollutant dynamics, and adaptive feature fusion to combine spatial and temporal representations. Meteorological variables, historical pollutant observations, temporal attributes, and station coordinates are incorporated to forecast particulate matter concentrations at multiple prediction horizons. A multi-task output layer jointly estimates future $PM_{2.5}$ concentrations and the probability of pollution-threshold exceedance, while quantile prediction provides an estimate of forecast uncertainty. The framework will be evaluated using publicly accessible air-quality observations and historical meteorological data, subject to verified data availability. Comparative experiments against persistence, conventional machine-learning, and temporal deep-learning baselines will assess forecasting accuracy, high-pollution event detection, robustness to missing observations, and spatial generalisation. The trained model will be deployed through a RESTful inference API and integrated into a weather-oriented application that presents pollution forecasts, emerging hotspots, uncertainty indicators, and location-specific environmental alerts. The study will also examine the contribution of wind-conditioned graph learning and adaptive fusion through controlled ablation experiments. By combining predictive modelling, spatial risk communication, and practical model deployment, MOSAIC-Air aims to establish a reproducible and resource-conscious framework for AI-assisted urban air-quality decision support.

Keywords: Air pollution forecasting, spatiotemporal deep learning, graph neural networks, wind-conditioned attention, temporal convolutional networks, adaptive feature fusion, hotspot detection, uncertainty estimation, environmental intelligence.